



UTM
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SECTION 2

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TUTORIAL ARTICLE

GROUP 17

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1. Main Problem

PT. MPM is a packaging manufacturing company. The company did not have a proper system to monitor and evaluate its business performance. As a result, top management struggled to make timely and accurate decisions because performance data was scattered, difficult to analyze and not presented in a meaningful way.

Several problems are identified:

- **Managing without visibility:** There was no single and centralized dashboard where leaders can log in to get a quick and complete company health check.
- **Data Everywhere, but No Insights:** Although sales numbers and production logs existed, but they were trapped in raw formats. It does not turn into actual and useful business insights.
- **Untracked Goals:** Management is difficult to track important health metrics (KPIs) such as whether actual sales were hitting their targets and forecasts, real-time revenue performance, total production capacity and line efficiency, inefficient decision making and reduce competitiveness against competitors.

2. Solution

To bridge the company's business insight, a BI dashboard was developed using the Waterfall approach. The process involved gathering raw data from different parts of the company, cleaning and processing it through ETL Process, and storing it in a data warehouse. The raw data was transformed into two interactive dashboards designed for everyday business tracking:

A. Sales Performance Dashboard

This dashboard gives leadership a clear view of commercial health by tracking:

- **Sales & Product Trends:** Depth breakdowns of how specific product categories are moving.
- **Revenue Streams:** Real time visibility into overall financial gains.
- **Targets vs. Reality:** Direct, side-by-side comparisons of actual sales figures against initial forecasts.
- **Client and Category Health:** Insights into which customer segments and product groupings generate the most value.

B. Production Performance Dashboard

This dashboard built for efficiency by allowing management to monitor manufacturing output closely:

- **Equipment Efficiency:** Tracks effectiveness across the manufacturing lines.
- **Operational Ratios:** Breaks down performance by quality ratio, product ratio, and time

utilization.

- **Output Tracking:** Contrasts planned production numbers against actual daily output.
- **Machine Health:** Monitors the status and downtime of individual machines to keep production steady.

Benefits

By shifting to centralized dashboards, top management are able to:

- **Gain centralized operational visibility:** Access operational and financial data from multiple departments through a single integrated dashboard, reducing data silos and providing a comprehensive view of organizational performance.
- **Monitor KPIs in real time:** Track key performance indicators (KPIs), such as sales performance, production output, and machine efficiency, continuously to support timely decision making and improve operational responsiveness.
- **Identify bottlenecks in real time:** Use visual analytics tools, including charts, heatmaps, and performance indicators, to quickly detect issues such as machine downtime, production inefficiencies and business areas before they become critical problems.
- **Support evidence based decision-making:** Rely on accurate and centralized data rather than intuition or delayed reports, enabling more informed and reliable strategic decisions.

3. User Stories

User Story 1 – Top Management

As a Top Manager,

I want to view company sales and production performance in one dashboard, so that I can make faster and more accurate business decisions.

User Story 2 – Production Manager

As a Production Manager,

I want to monitor OEE, machine performance, and production achievement, so that I can identify production inefficiencies and improve productivity.

User Story 3 – Sales Manager

As a Sales Manager,

I want to compare actual sales with forecasted sales, so that I can evaluate sales performance and improve future planning.

User Story 4 – Data Analyst

As a Data Analyst,
I want to build a clean, automated ETL pipeline from our raw reports into a centralized Google Sheets data warehouse,
so that the management dashboards always display accurate, standardized, and real time information.

User Story 5 – Operations Manager

As an Operations Manager,
I want to filter dashboard data by date range, quarter, customer ID, product category, product division, and machine,
so that I can easily isolate bottlenecks and analyze performance trends from different departmental perspectives.

4. ERD Diagram

Figure 1 shows the Entity Relationship Diagram (ERD) for the PT. MPM Performance Monitoring Business Intelligence system. The ERD is designed using UML class diagram notation to represent the main entities, attributes, primary keys, foreign keys, and relationships involved in the system.

The ERD consists of two sections. The first one is the access management module of the dashboard, containing Department, SystemUser, Dashboard, and DashboardAccess. This module determines users' access rights and permissions to the BI dashboards according to their role and department.

The second part is the operational performance data module, containing Customer, ProductCategory, Product, SalesTransaction, SalesForecast, InventoryStock, Machine, ProductionReport, and ProductionKPI. These entities are used to store sales, production, inventory, and machine performance data that will be used to create the sales performance dashboard and production performance dashboard.

In this design, a customer can have many sales transactions and sales forecasts, while a product can be involved in many sales transactions, forecasts, inventory records, and production reports. Production reports are connected to a machine and include production KPI values such as availability performance, quality, and Overall Equipment Effectiveness (OEE). For this reason, the ERD is used to assist the BI dashboard in structuring the data required for monitoring PT. MPM's sales and manufacturing performance.

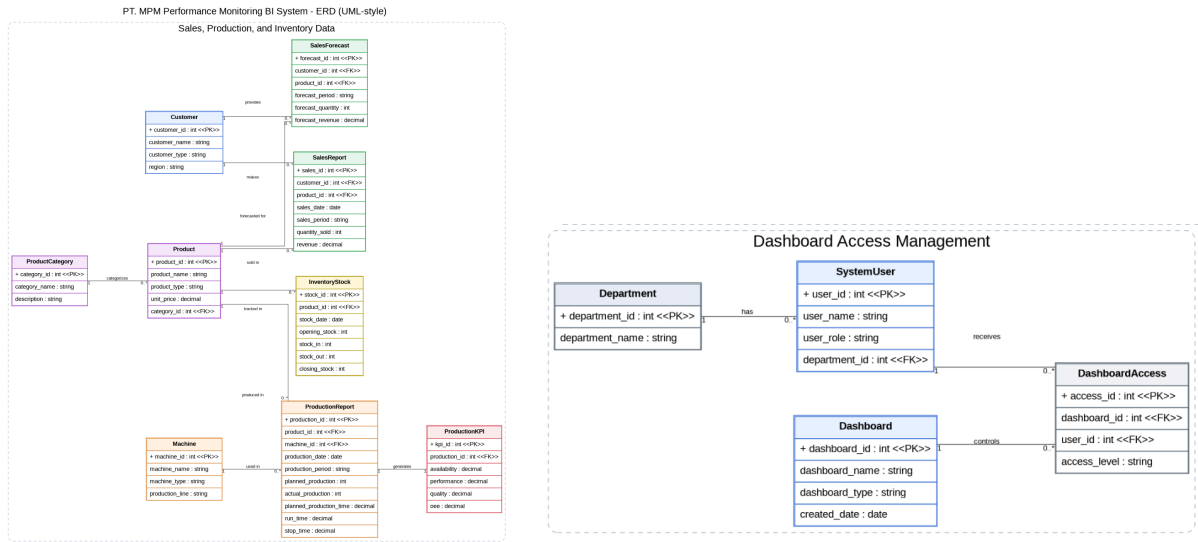


Figure 1: ERD of PT. MPM Performance Monitoring BI System

5. Use Case Diagram

The use case diagram of PT. MPM Performance Monitoring BI System is shown in Figure 2. The diagram was created based on the user stories identified in the case study. It shows the different ways users use the BI dashboard system based on their duties. It demonstrates the various ways users interact with the BI dashboard system based on their assigned duties.

Top Management, Production Manager, Sales Manager, Data Analyst, Operations Manager are the main actors of the system. Top Management/Company Executive can use the dashboard to monitor the overall performance of the company to support business decision-making. The Production Manager will view actual OEE, machine performance, and production achievement, while the Sales Manager will compare actual sales with forecasted sales and review sales performance.

The Data Analyst will be tasked with obtaining the raw reports, cleaning and transforming the data, loading the data into the Google Sheets data warehouse, and creating the BI dashboard. The Operations Manager can filter the dashboard to analyze performance within a date range, quarter, customer, product category, product division, and machine basis. The overall use case diagram illustrates the support that the BI system provides in terms of data preparation, dashboard visualization, performance monitoring, and decision-making.

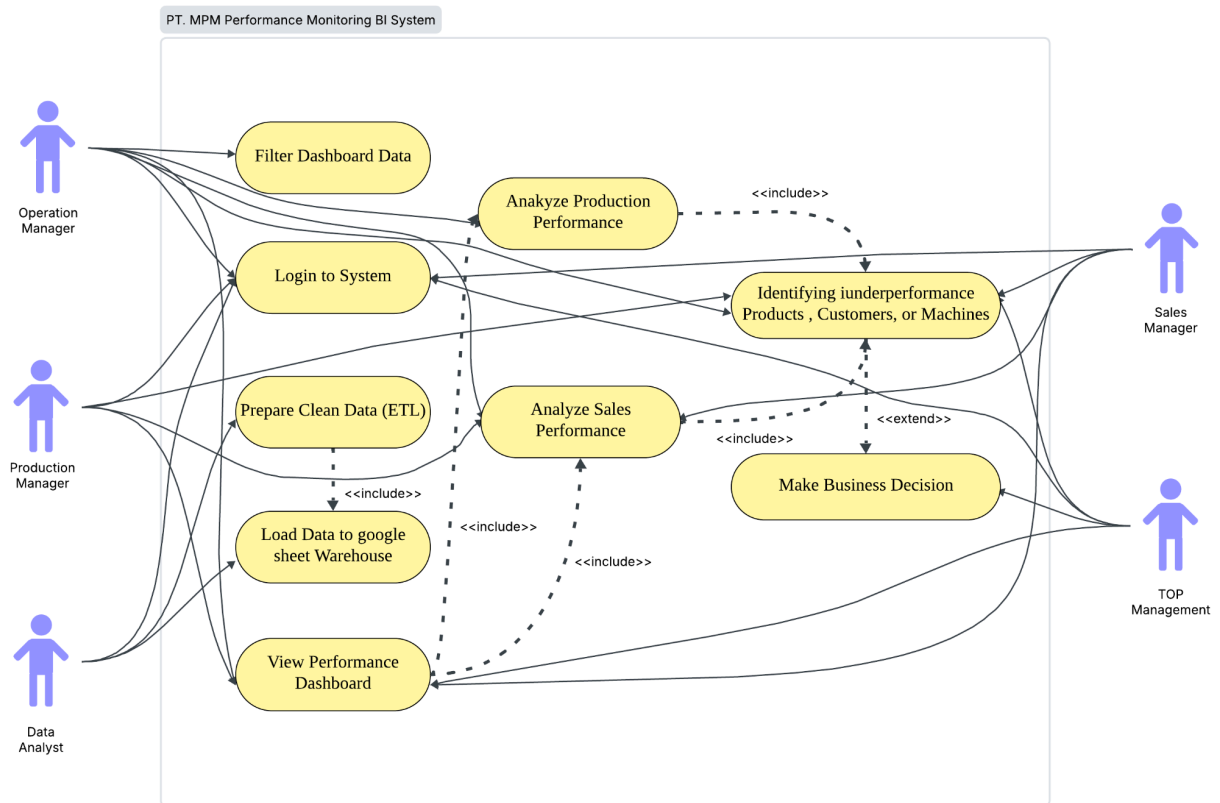
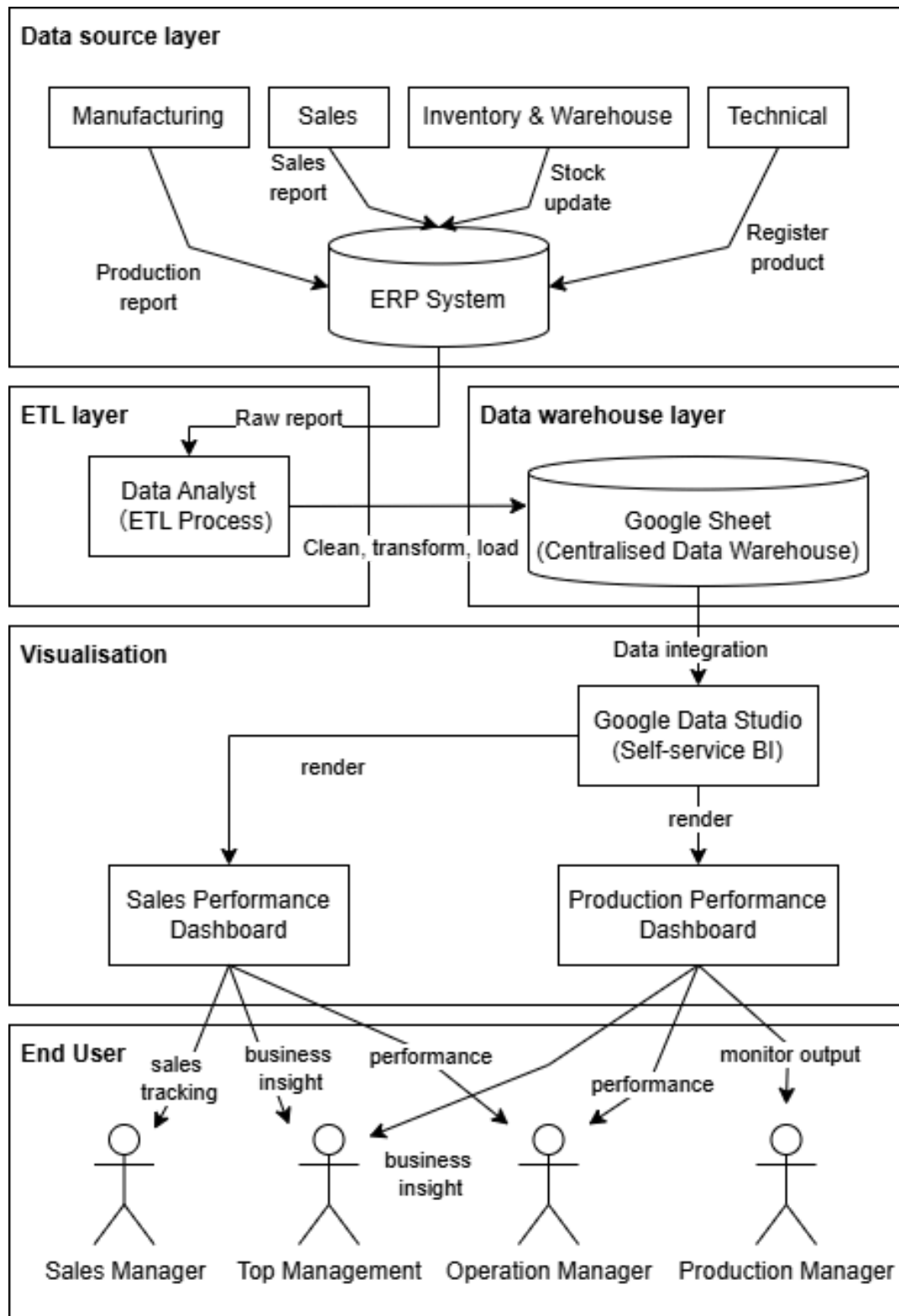


Figure 2: Use Case Diagram of PT. MPM Performance Monitoring BI System

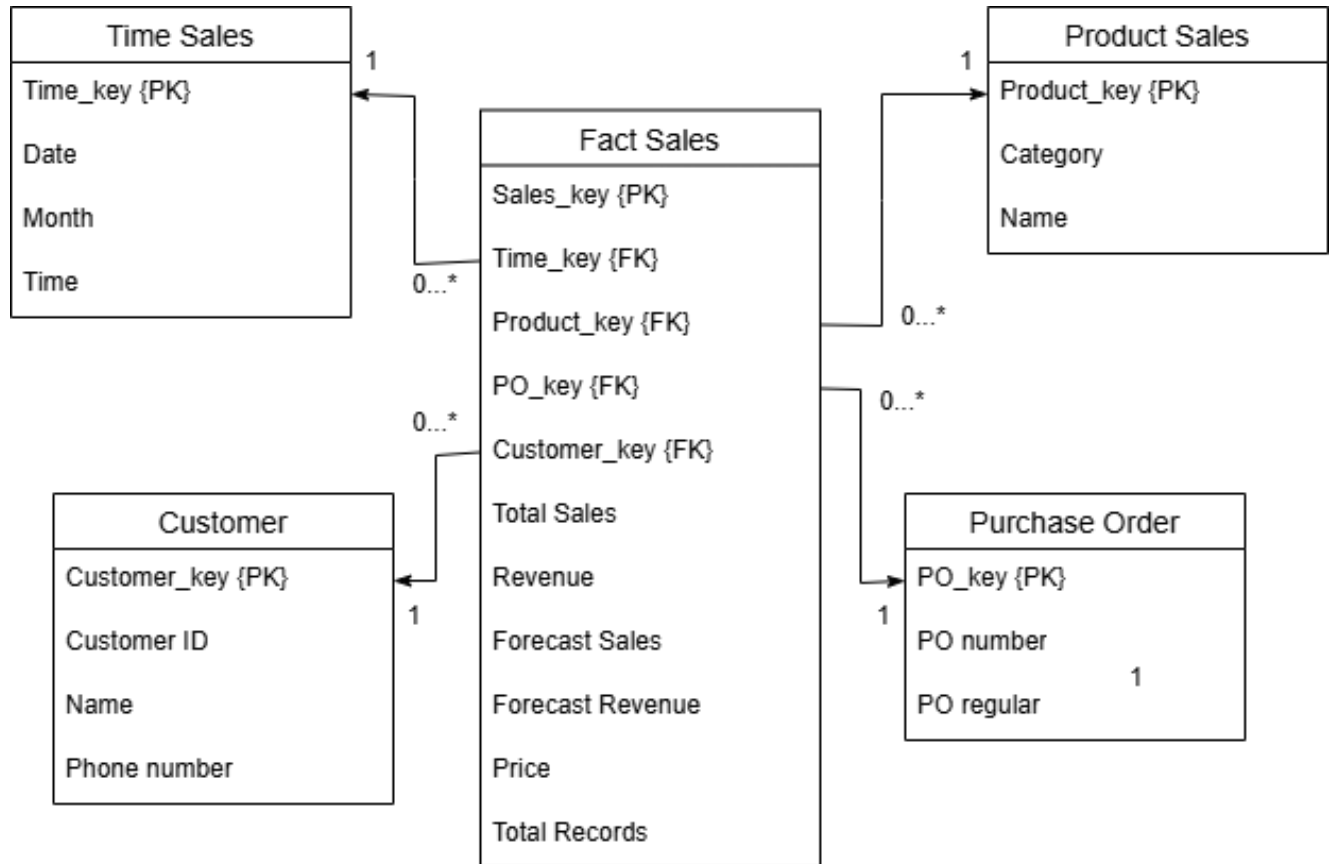
System Architecture



The system architecture is designed to solve the problems faced by the company which is the integration of distributed raw performance data into a unified business intelligence process. The architecture is structured into five layers.

Layers	Description
Data source	Collect original department data (manufacturing, sales, inventory, technology) through the ERP system.
ETL layer	Data analysts are responsible for managing automated data pipelines. Here, the raw report is extracted directly from the ERP system, and then the data is cleaned to delete irrelevant information and converted into a structured format like converting the date string format to the correct date format.
Data warehouse layer	Load the cleaned and structured data into Google Sheets, which serves as the centralised data repository of the organisation. This creates a single source of truth for all performance data.
Visualisation	The Google Form Data Warehouse is connected to Google Data Studio, which is a self-service business intelligence tool. The platform provides two main interactive user interfaces, sales performance dashboard and production performance dashboard.
End user access	Key stakeholders can access the final dashboard, including top management, production managers, sales managers and operations managers. This level allows the management team to view key business operations on a single screen, track key performance indicators and make decisions based on accurate data.

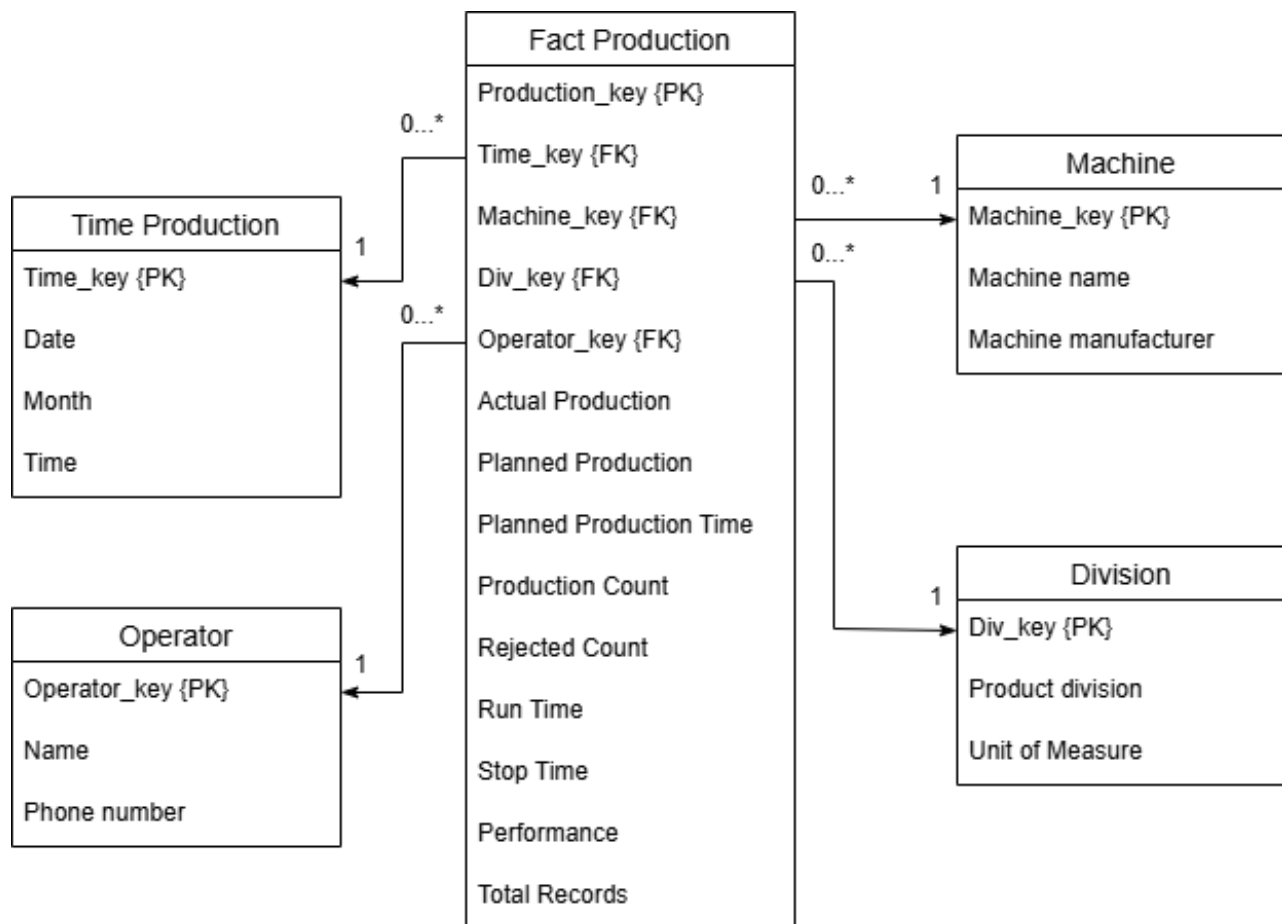
Multidimensional Design



Sales Performance

This schema is designed to track the business's health status, monitor the revenue, and compare the actual sales data with the initial predictions.

Table	Description
Fact table	This central fact table stores the quantitative data collected from the sales activities.
Dimension table	The fact table is connected to multiple descriptive dimension tables to let the users conduct filtering and analysis of the data. This structure allows the sales and operations managers to analyse performance trends by specific customer groups or product categories.



Production Performance

This schema is designed to monitor manufacturing output, track equipment efficiency, and check daily production operations.

Table	Description
Fact table	This central fact table contains all the key performance indicators related to manufacturing efficiency.
Dimension table	The fact table is connected to multiple descriptive dimension tables to assist the management in identifying specific operational bottlenecks. This allows the production manager to break down the performance and

	monitor the status and downtime of each machine.
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